$$D[i,j] = D[i,k] + D[k,j]$$

$$D^0 = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & \infty \\ \infty & 7 & 0 & 1 \\ 6 & \infty & \infty & 0 \end{bmatrix} \end{array}$$

$$D^1 = k = a$$

$$\begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & 5 & \infty \\ \infty & 9 & 0 & 1 \\ 6 & \infty & 9 & 0 \end{bmatrix} \end{array}$$

$$D[b,c] = D[b,a] + D[a,c]$$
$$= 2 + 3 = 5$$

$$D[b,d] = D[b,a] + D[a,d]$$
$$= 2 + \infty = \infty$$

$$D[c,b] = D[c,a] + D[a,b]$$
$$= \infty + \infty = \infty$$

$$D[c,d] = D[c,a] + D[a,d]$$
$$= \infty + 6 = \infty$$

$D[d,b] \Rightarrow D[d,a] + D[\cancel{d},b]$

$= 6 + \infty = \infty$

$D[d,c] = D[d,a] + D[a,c]$

$= 6 + 3 = 9$

$D^2 = \cancel{k} = b$

$$\begin{array}{c|cccc} & a & b & c & d \\ \hline a & & & & \\ b & & & & \\ c & & & & \\ d & & & & \end{array} \quad \begin{array}{cc} & \\ \begin{array}{c} a \\ b \\ c \\ d \end{array} & \begin{bmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & 5 & \infty \\ 9 & \infty & 0 & 1 \\ 6 & \infty & 9 & 0 \end{bmatrix} \end{array}$$

$D[a,c] = D[a,b] + D[b,c]$

$= \infty + 5 = \infty$ 3 is min

$D[a,d] = D[a,b] + D[b,d]$

$= \infty + \infty = \infty$

$D[c,a] = D[c,b] + D[b,a]$

$= \cancel{\infty}\,7 + 2 = \cancel{\infty}\,9$

$D[c,d] = D[c,b] + D[b,d]$

$= 7 + \infty = \infty$ 1 is min

$D[d,a] = D[d,b] + D[b,a]$

$= \infty + 2 = \infty$ 6 is min

$D[d,c] = D[d,b] + D[b,c]$

$= \infty + 5 = \infty$.   9 is min

$D^3 = k = c$

$$D^3 = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 10 & 3 & 4 \\ 2 & 0 & 5 & 6 \\ 7 & 7 & 0 & 1 \\ 6 & \infty & 9 & 0 \end{bmatrix} \end{array}$$

$D[a,b] = D[a,c] + D[c,b]$

$= 3 + 7 = 10$

$D[a,d] = D[a,c] + D[c,d]$

$= 3 + 1 = 4$

$D[b,a] = D[b,c] + D[c,a]$

$= \infty + \infty = \infty$

$D[b,d] = D[b,c] + D[c,d]$

$= \overset{5}{\cancel{\infty}} + 1 = \cancel{\infty}\ 6$

$D[d,a] = D[d,c] + D[c,a]$

$= \cancel{9+9}\ \infty + \infty = \infty > 6$

$D[d,b] = D[d,c] + D[c,b]$

$= \infty + 1 = \infty$

$D^4 = k = d$

$$D = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 10 & 3 & 4 \\ 2 & 0 & 5 & \infty \\ 2 & 7 & 0 & 1 \\ 6 & \infty & 9 & 0 \end{bmatrix} \end{array}$$

$$D[a,b] = D[a,d] + D[d,b]$$
$$= \infty + \infty = \infty > 10$$
$$D[a,c] = D[a,d] + D[d,c]$$
$$= \infty + \infty = \infty > 3$$
$$D[b,a] = D[b,d] + D[d,a]$$
$$= \infty + 6 = \infty > 3$$
$$D[b,c] = D[b,d] + D[d,c]$$
$$= \infty + \infty = \infty > 5$$
$$D[c,a] = D[c,d] + D[d,a]$$
$$= 1 + 6 = 7$$
$$D[c,b] = D[c,d] + D[d,b]$$
$$= 1 + \infty = \infty \geq 7$$